LOAN APPROVAL PREDICTION USING RANDOM FOREST

Machine Learning Mini Project | Student: P. Rehan Khan

1. Problem Statement

The objective of this project is to build a classification model that predicts whether a loan application will be **Approved** or **Rejected** based on applicant details such as income, loan amount, CIBIL score, employment status and asset values.

2. Dataset Description

Dataset: **Loan Approval Prediction Dataset** (Kaggle, Archit Sharma). It contains **4,269 records and 13 columns**, including 11 predictor features, a loan identifier and the target variable *loan_status*. The identifier *loan_id* was excluded from model inputs.

Source: https://www.kaggle.com/datasets/architsharma01/loan-approval-prediction-dataset

3. Data Preprocessing

The dataset was explored using Pandas. Missing values and duplicate rows were checked; none were found. Leading/trailing spaces were removed from categorical values. The columns *education*, *self_employed* and *loan_status* were converted into numeric values using dictionary mapping. The data was separated into input features (X) and target (y), then split into 80% training data (3,415 records) and 20% testing data (854 records) using random_state=42.

4. Model Used

A **RandomForestClassifier** with 100 decision trees (n_estimators=100) and random_state=42 was trained on the training set. The trained model was used to predict loan status for the test set.

5. Model Evaluation

Metric	Result
Accuracy	98.01%
Precision	98.14%
Recall	98.69%
F1-score	98.42%

Confusion matrix (actual rows; predicted columns): [[308, 10], [7, 529]]. The model correctly classified 837 out of 854 test records.

6. Feature Importance and Key Finding

CIBIL score was the most influential feature in the trained Random Forest model, with an importance score of **0.8119**. It was followed by loan term (0.0647) and loan amount (0.0281). Education had the lowest importance score (0.0022). These are model-derived importance scores, not causal effects or direct percentages of loan approval.

7. Conclusion

The Random Forest classifier achieved 98.01% accuracy on the held-out test set and showed strong precision, recall and F1-score. Feature importance identified CIBIL score as the dominant predictor among the input features. The model demonstrates how applicant and financial information can be used for loan-status prediction; its results should be interpreted as performance on this dataset and test split.